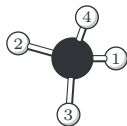


rigid methane



this paper, by parity

Albert et al., Table II

A_1^-, A_2^+

5 each

(A, A, 1): 5

E^-, E^+

1 each

(¹E, ²E, 1): 1

(²E, ¹E, 1): 1

T_2^-, T_1^+

3 each

(T, T, 3): 3

restriction $T_d \rightarrow T$: $A_{1,2} \rightarrow A$, $E \rightarrow {}^1E \oplus {}^2E$, $T_{1,2} \rightarrow T$; parity is ours alone

ℓ_g lifts to $r \mapsto rg$; a species sees g through $\Gamma(g)$

monodromy group $T/\ker \Gamma$:

$A \mapsto 1$, ${}^1E, {}^2E \mapsto C_3$, $T \mapsto T$

the T state is the invariant of $T \otimes T$:

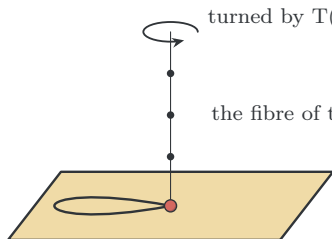
Schmidt rank 3; 9 of the 16 spin states

turned by $T(g)$

the fibre of the T isomer, $\mathfrak{d} = 3$

$SO(3)/T$

ℓ_g



$H = T_d(M) \cong S_4$
even relabellings; odd ones starred
 $H_{\text{rot}} = T \cong A_4$, twelve rotations

spin $16 = 5 A_1 \oplus E \oplus 3 T_2 (H)$
 $= 5 A \oplus {}^1E \oplus {}^2E \oplus 3 T (H_{\text{rot}})$